

CodeAlpha C++ Internship

TASK 3: Sudoku Solver

Complete source code for this internship task.

```
#include <iostream>
using namespace std;

const int N = 9;

void printGrid(int grid[N][N]) {
    for (int row = 0; row < N; row++) {
        for (int col = 0; col < N; col++)
            cout << grid[row][col] << " ";
        cout << '\n';
    }
}

bool isSafe(int grid[N][N], int row, int col, int num) {
    for (int x = 0; x < N; x++)
        if (grid[row][x] == num)
            return false;

    for (int x = 0; x < N; x++)
        if (grid[x][col] == num)
            return false;

    int startRow = row - row % 3;
    int startCol = col - col % 3;

    for (int i = 0; i < 3; i++)
        for (int j = 0; j < 3; j++)
            if (grid[startRow + i][startCol + j] == num)
                return false;

    return true;
}

bool solveSudoku(int grid[N][N]) {
    for (int row = 0; row < N; row++) {
        for (int col = 0; col < N; col++) {

            if (grid[row][col] == 0) {
                for (int num = 1; num <= 9; num++) {

                    if (isSafe(grid, row, col, num)) {
                        grid[row][col] = num;

                        if (solveSudoku(grid))
                            return true;

                        grid[row][col] = 0;
                    }
                }
                return false;
            }
        }
    }

    return true;
}

int main() {
    int grid[N][N] = {
        {5,3,0,0,7,0,0,0,0},
        {6,0,0,1,9,5,0,0,0},
        {0,9,8,0,0,0,6,0,0},
        {8,0,0,0,6,0,0,0,3},
        {4,0,0,8,0,3,0,0,1},
        {7,0,0,0,2,0,0,0,6},
        {0,6,0,0,0,0,2,8,0},
        {0,0,0,4,1,9,0,0,5},
        {0,0,0,0,8,0,0,7,9}
    };

    cout << "Original Sudoku:\n";
    printGrid(grid);

    if (solveSudoku(grid)) {
        cout << "\nSolved Sudoku:\n";
        printGrid(grid);
    } else {

```

```
        cout << "\nNo solution exists.\n";  
    }  
    return 0;  
}
```